

Technical Portfolio

Project: AI-Powered Quality Engineering (QE) Agent System

Project Overview

Designed and developed an AI-powered Quality Engineering workflow platform that automates the end-to-end software testing lifecycle. The solution transforms product artifacts such as PRDs, API Specifications, and User Stories into test plans, generated test cases, execution results, and defect triage reports through a multi-agent architecture. The platform supports both CLI and REST API-based execution with Human-in-the-Loop (HITL) governance controls.

Key Technologies & Tools

- Java 17
- Maven
- REST API / HTTP Server
- JSON
- TestNG
- OpenAI GPT-4o Mini / OpenRouter
- JSON Processing
- JDK HttpServer
- OWASP Security Practices
- Git & GitHub
- Prompt Injection Detection
- Prompt Engineering
- AI Agent Architecture
- Code Sanitization Framework
- Human-in-the-Loop (HITL) Workflow Controls
- Parallel Test Execution Framework

Solution Components

Component	Description
Test Planner Agent	Generates risk-based test plans from PRDs, API specifications, and user stories

Test Generator Agent	Produces executable TestNG test scripts and test data	
Test Executor Agent	Executes tests in parallel with retry, timeout, and flakiness detection	
Defect Triage Agent	Performs defect clustering, root cause analysis, severity and priority assignment	
QE Workflow Orchestrator	Coordinates all agent activities across the testing lifecycle	
Security Layer	Performs prompt injection detection and code sanitization	
Backend Server	Exposes workflow through REST endpoints for UI integration	

Role

AI Engineer / Quality Engineering Automation Developer

Key Responsibilities

- Designed and implemented AI agents for Test Planning, Test Generation, Test Execution, and Defect Triage.
- Integrated LLMs to automate test artifact creation from business requirements and API specifications.
- Developed REST APIs to expose AI-powered testing workflows.
- Implemented test automation, security validation, and defect triage capabilities.
- Developed prompt injection detection and code sanitization mechanisms.
- Built automated defect analysis and root cause identification capabilities.
- Enabled parallel test execution with retry and flaky-test detection mechanisms.
- Configured Human-in-the-Loop review and approval processes.
- Performed system architecture design, testing, and deployment support.

Outcome

- Automated end-to-end quality engineering workflows.
- Reduced manual effort in test planning and test case creation.
- Improved testing efficiency through AI-generated test scenarios.
- Accelerated defect identification and prioritization.
- Increased test coverage and quality assurance effectiveness.

- Enhanced platform security through AI governance and validation controls.

Business Benefits

- Faster software delivery through intelligent test automation.
- Improved defect detection and resolution turnaround time.
- Reduced QA operational costs through automation.
- Increased software reliability and product quality.
- Better governance and compliance with human approval checkpoints.
- Scalable AI-powered testing framework supporting future project growth.

Key Achievements

- Implemented a complete 4-stage AI-driven Quality Engineering workflow.
- Achieved automated defect clustering and root-cause identification.
- Integrated secure AI model invocation with fallback processing.
- Developed REST APIs supporting frontend and backend interoperability.
- Built reusable and scalable agent architecture for future extensibility.